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Quantitative consequence analysis using Sedov-Taylor blast wave model. Part I: Model description and validation

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ABSTRACT

BLEVE (Boiling Liquid Expanding Vapor Explosion) phenomenon is one of the major industrial accidents observed in gas processing industry, which remains a major concern for risk decision-makers. BLEVE blast wave mechanism has been widely studied by several authors who proposed simplified approaches based on simple physical models or empirical correlations, but only few approaches including analytical solutions have been undertaken. Moreover, the simplified and empirical approaches are not very satisfactory because they overestimate overpressure measures. In this paper (Part I), an analytical model based on Sedov-Taylor blast wave solution and self-similar theory, which is of great interest in various fields of physics, is proposed for estimating BLEVE overpressure effects. The parameters characterizing the blast wave evolution (overpressure, radius and velocity) are established by applying the Vashy-Buckingham theorem (Pi theorem). To demonstrate the ability of the proposed model to deliver reliable predictions, a validation with large and medium-scale BLEVE experiments issued from the literature is carried out. Furthermore, a comparison with the TNT equivalent model and other models (empirical and simplified physical) is performed. The results of these comparisons are very encouraging and show good agreement in terms of precision.

This is Part I of two papers, focusing on description and validation of the Sedov-Taylor blast wave model. Part II deals with application of the model on a LPG accumulator in an Algerian gas processing unit.

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1. Introduction

The process industries involve not only the performance of production systems in terms of quality and reliability, but also in terms of safety. The prevention of major industrial accidents with severe consequences on people, equipment and environment (such as fires, explosions, BLEVEs, toxic releases, etc.) remains a major concern of decision-makers and industrial experts (CCPS, 2010).

BLEVE (Boiling liquid expanding vapor explosion) is one of the major consequences in process accidents which have been observed mostly with liquefied petroleum gas (LPG) vessels in gas processing industries. Their effects on the environment generally appear in three ways: blast overpressure, radiation from the fireball if the liquid involved is flammable and fragment hazards. Therefore, it is important to estimate the potential damage that would be caused by such explosion.

BLEVE is a complex phenomenon resulting from many factors that are difficult to summarize in mathematical equations or models. However, BLEVE mechanism and overpressure modeling have been widely studied by several authors who proposed simplified approaches based on simple physical models or empirical correlations (e.g. TNO procedure from the Yellow book and TNT equivalent curves) which use the scaled distance curves to calculate the overpressure (Abbasi et al., 2007; Baker et al., 1983; Birk et al., 2007; Bubbico and Marchini, 2008; Casal and Salla, 2006; Genova et al., 2008; Kielec and Birk, 1997; Papazoglou and Aneziris, 1999; Planas-Cuchi et al., 2004; Prugh, 1991; Salla et al., 2006; Strehlow and Baker, 1976; Tang and Baker, 2000; TNO, 2005). Such approaches are not very satisfactory because they usually include an exper-

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imentally adjusted reduction factor and mostly overestimate the blast overpressures (Laboureur et al., 2015, 2014; Van Den Berg et al., 2004).

In addition, the blast wave resulting from the hydrocarbon explosion can also be predicted by computational fluid dynamics codes (FLACS software for example) which are touted as more fundamental than other approaches. However, these CFD codes are limited to deflagration cases and cannot be extended to detonation models (Bang et al., 2016). For this reason, TNO organization notes that the CFD approach is not necessarily more fundamental than other simple physical approaches (CCPS, 2010; TNO, 2005).

Self-similar theory of strong explosions can be used to go deeper in the understanding of the physical phenomenon leading to the BLEVE based on Sedov-Taylor model where the evolution of radius, velocity and blast overpressure can be predicted for a large quantity of exploding hydrocarbon energy (Bang et al., 2016; Batcherlor, 1963; Caillard et al., 2004; Gordon et al., 2013; Landau and Lifshitz, 1987; Liberman, 2008; Sedov, 1971, 1959, Taylor, 1950a,b).

In this Part I, an analytical modeling approach based on Sedov-Taylor model is proposed and validated with experimental data issued from British Gas tests (Johnson et al., 1990) and Birk et al. (2007) experiments. By applying the Vashy-Buckingham theorem (Buckingham, 1914), the physical quantities characterizing the evolution of the shock wave are determined. Furthermore, a comparison with the TNT equivalent model, empirical and simplified physical models is performed.

This article is organized as follows: Section 2 deals with the BLEVE overpressure modeling using the proposed Sedov-Taylor model. Section 3 focuses on validating the model with the TNT equivalent model based on large and medium-scale experiments. Section 4 is dedicated to the concluding remarks.

2. BLEVE overpressure modeling

2.1. Overview on the BLEVE phenomenon

In the petroleum and chemical industry, it is common to resort to gas liquefaction. This process allows a larger quantity of product to be stored or transported. Liquefaction requires lowering the temperature of the gas below its boiling point (liquefaction by cooling) and/or increasing its pressure to go beyond its saturation pressure (liquefaction by compression). One of the major disadvantages of this storage mode is BLEVE risk which corresponds to a violent vaporization of explosive nature following the rupture (loss of confinement) of a reservoir containing a liquid at a temperature significantly higher than its normal boiling point at atmospheric pressure (CCPS, 2010; Champassith, 2014).

A simple accident analysis can demonstrate the potential destruction of a BLEVE. Thus, the 80-odd major BLEVEs that have occurred between 1940 and 2005 have costed more than 1000 lives and have injured more than 10000 persons in addition to harming property worth billions of dollars (Abbasi et al., 2007).

Several causes of BLEVE are identified, including: thermal stresses, fatigue, corrosion, poor manufacture, pressure stresses and reduction in material strength due to high wall temperatures (Birk et al., 2007).

The BLEVE theory involves the notion of superheat limit temperature (SLT). When a liquid heats up at a given pressure, its temperature increases until reaching the boiling point, so that vapor bubbles develop on the active sites. When there are not enough nucleation sites in the liquid, the boiling point can be exceeded without boiling. In this case, the liquid is said to be superheated. However, there is a temperature limit at a given pressure beyond which vapor bubbles develop throughout the liquid, even in the absence of nucleation sites. This limit is the superheating limit temperature (Casal and Salla, 2006; Salla et al., 2006).

The effects of BLEVE on the environment generally appear in three ways (INERIS, 2002; Török et al., 2011):

- Propagation of overpressure wave: This is the consequence of an explosion which results in the propagation of a pressure wave in atmosphere at the supersonic speed (> 340 m/s in air at 15 °C) at the start.
- Projection of fragments: The impact of a projectile is likely to produce two types of effects on people and/or structures: (a) shock, which can cause harm to human body and structures, (b) penetration, whether in human body or in structures.
- Thermal effect: With flammable liquid, the fireball may become preponderant in terms of consequences. However, the observed effects depend on the exposure time as follows: (a) when the phenomenon exceeds 2 min (steady state), the effect distances are calculated as a flux expressed in kW/m^2, this is the case for pool fires and jet fires; (b) when the duration of the phenomenon is less than 2 min (dynamic regime), the distances are calculated in terms of received thermal doses expressed in $[(kW/m^2)^{4/3}]\cdot s$, this is the case of a BLEVE or a Boil-Over where the duration of exposure is very short.

With the goal of controlling and reducing industrial risks, understanding and characterizing BLEVE effects are a crucial problem. Although it was the subject of several researches and publications, our understanding of the different mechanisms of BLEVE remains insufficient, there is not a definitive and unique method for modeling these effects. The problem is more complex since the generated effects are of different natures (mechanical, thermal, etc.) and refer to specific modeling methods. This Part I focuses on one specific hazard of BLEVE: the blast wave.

2.2. Fundamentals of self-similar theory: Sedov-Taylor model

The different factors which may influence the propagation of an explosion shock wave severely hamper the theoretical treatment of the problem. To arrive at a simple solution of this problem, Taylor (1950b) and Sedov (1971, 1959) assumed that the source of an explosion to be a point. Their solution is so-called self-similar theory, which describes the evolution of the blast wave arising from a strong explosion based on exact analytical expressions.

Sedov-Taylor blast wave model is of great interest in various areas of physics. It is applied for example in astrophysics studies (Dale, 2015), nuclear explosion modeling (Taylor, 1950a), plasma studies (Caillard et al., 2004), munitions explosions (Gordon et al., 2013), thermo-hydrodynamics studies (Kwak et al., 2012), and recently in hydrogen pipeline explosion modeling (Bang et al., 2016). This model with a spherical geometry is perfectly described in fluid mechanics book of Landau and Lifshitz (1987), other geometries (cylindrical and plane) are studied by Sedov (1959).

Let us consider an ideal gas medium of density ρ_0 ($\rho_0 = 1.22\ kg/m^3$ for air at 15 °C) and specific heats ratio γ at ambient pressure P_0, in which an energy E is suddenly released in an infinitely concentrated form during a short time interval. This initial energy produces a spherical shock wave whose radius R (from the center) is a function of the time t, as shown in Fig. 1 (Batcherlor, 1963; Kandula and Freeman, 2008; Liberman, 2008).

Dimensionless analysis is a powerful approach for estimating the shock wave parameters (radius, velocity and overpressure). In this subsection, dimensional analysis based on the Pi theorem of Vaschy-Buckingham is used to determine a single formula describing the full-time behavior of the explosion from the beginning (Buckingham, 1914; de Izarra et al., 2002).

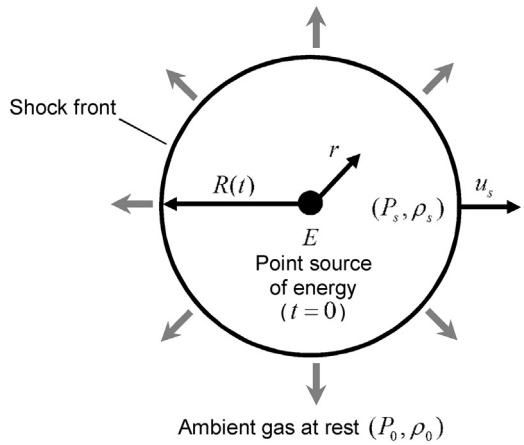

Fig. 1. Schematic of a point source explosion.

Table 1
Dimensions of the principal quantities.

Physical quantity	Notation	Fundamental units
Energy	E	ML^2T^{-2}
Density of the gas at rest	ρ_0	ML^{-3}
Radius of the blast wave front	R	L
Time	t	T

2.2.1. Dimensional analysis: Application of the Pi Theorem

Most of the time, the overpressure P_S just behind the spherical discontinuity of the radius R produced by the explosion is very large compared to the ambient pressure P_0 of the medium at rest. This assumption is fully justified because the internal energy of the ideal gas at rest is negligible compared to the initial energy value E (strong explosion). Indeed, the only parameters of the problem are then E and ρ_0, and all the quantities appearing in the solution can only be a combination of the time t, the density ρ_0 and the energy E. We can therefore use dimensional analysis to identify some of the expected dependencies. In the following, the Pi theorem is applied with four physical quantities listed in Table 1, where L, M and T denote length, mass and time, respectively.

The Pi theorem indicates that, with four dimensionally independent physical quantities and three fundamental units, only one dimensionless $\prod$ can be obtained. Since we are interested in the time variation of the shock wave front radius, the dimensionless $\prod$ for radius R can be given by:

$$\prod_R = R^1 E^a \rho_0^b t^c \tag{1}$$

where a, b and c are easily found by considering that $\prod_R$ is a dimensionless factor ($\left[\prod_R\right] = 1$). We obtain:

$$\prod_R = RE^{-1/5}\rho_0^{1/5}t^{-2/5} = R\left(\frac{\rho_0}{E}\right)^{1/5} t^{-2/5} \tag{2}$$

The relation giving the evolution of the shock wave front as a function of time is:

$$\prod_R = \beta \tag{3}$$

where β is a dimensionless constant (only depending on the specific heats ratio γ), which can be found by solving the motion equations.

From Eq. (2) and Eq. (3) we can obtain the well-known result of the self similar blast wave model, with an evolution of R as a function the time t as (Caillard et al., 2004; Sedov, 1959; Taylor, 1950b):

$$R(t) = \beta\left(\frac{Et^2}{\rho_0}\right)^{1/5} \tag{4}$$

The propagation velocity of the shock wave is given by Eq. (5).

$$u_s = \frac{dR}{dt} = \frac{2R}{5t} = \beta\frac{2}{5}\left(\frac{E}{\rho_0 t^3}\right)^{1/5} = \frac{2}{5}\beta^{5/2}\left(\frac{E}{\rho_0}\right)^{1/2} R^{-3/2} \tag{5}$$

The constant β is determined by the following condition which states that the total energy in the zone bounded inside the blast wave sphere remains constant and equal to the energy E released in the explosion (Kandula and Freeman, 2008; Landau and Lifshitz, 1987):

$$E = \int_0^R \rho\left(\frac{u^2}{2} + \frac{c_s^2}{\gamma(\gamma-1)}\right) 4\pi r^2 dr = 4\pi\int_0^R \left(\frac{1}{2}\rho u^2\right) r^2 dr + 4\pi\int_0^R \frac{P}{\gamma-1} r^2 dr \tag{6}$$

where c_s is the velocity of sound (which is expressed by $c_s = \sqrt{\gamma P/\rho}$) and r is the radial coordinate. In the case of a diatomic ideal gas such as air, for which $\gamma = 1.4$, the value of β is 1.03.

Thus, using the approximations presented by Liberman (2008), Eq. (4) becomes:

$$R(t) = \left(\frac{75}{16\pi}\frac{(\gamma-1)(\gamma+1)^2}{3\gamma-1}\right)^{1/5}\left(\frac{Et^2}{\rho_0}\right)^{1/5} \tag{7}$$

This approximation is quite close to the exact solution of the problem.

2.2.2. Rankine-Hugoniot relations

The shock wave overpressure can be expressed using the Rankine-Hugoniot relations for strong shock. These relations serve as boundary conditions of the shock wave front for Sedov-Taylor blast model solution. They can be expressed across a shock of finite strength at $r = R(t)$ as (Kandula and Freeman, 2008; Sedov, 1959; Taylor, 1950b):

$$\frac{\rho_s}{\rho_0} = \frac{\gamma+1}{\gamma-1}\left[1 + \frac{2}{\gamma-1}\left(\frac{c_{s0}}{u_s}\right)^2\right]^{-1}, \quad \frac{P_s}{P_0} = \frac{2\gamma}{\gamma+1}\left(\frac{u_s^2}{c_{s0}^2}\right) - \frac{\gamma-1}{\gamma+1} \tag{8}$$

where ρ_s and P_s denote respectively the density and pressure of the gas behind the shock wave. c_{s0} is the speed of sound in the undisturbed medium (which is expressed by $c_{s0} = \sqrt{\gamma P_0/\rho_0}$). The shock velocity u_s (relative to the undisturbed gas) is defined by Eq. (5).

In the strong shock condition when $P_s/P_0 \to \infty$ ($P_s/P_0 \gg 1$), Eq. (8) reduces to:

$$\rho_s = \rho_0\frac{\gamma+1}{\gamma-1}, \quad P_s = \frac{2\gamma}{\gamma+1}\left(\frac{u_s^2}{c_{s0}^2}\right)P_0 = \frac{2}{\gamma+1}\rho_0 u_s^2 \tag{9}$$

Substituting u_s from Eq. (5) in Eq. (9), we obtain the velocity of shock wave propagation as:

$$P_s = \frac{2}{\gamma+1}\rho_0 {u_s}^2 = \frac{8\beta^2}{25(\gamma+1)}{\rho_0}^{3/5}E^{2/5}t^{-6/5} = \frac{8\beta^5}{25(\gamma+1)}\frac{E}{R^3} \tag{10}$$

It is easily seen from Eq. (10) that the overpressure decreases with time as:

$$P_s \propto \rho_0 {u_s}^2 \propto \rho_0 \left(\frac{E}{\rho_0}\right)^{2/5} \propto \frac{E}{R^3} \tag{11}$$

2.3. Explosion energy estimate

Various models have been proposed in literature to estimate the BLEVE explosion energy using expansion energy notion which is based on the change in thermodynamic states of the product stored in the tank (Birk et al., 2007; Casal and Salla, 2006; Genova et al., 2008; Planas-Cuchi et al., 2004; Prugh, 1991; Roberts, 2000; TNO, 2005). Recently, Laboureur et al. (2015, 2014) have carried out a comparison of these models with multi-scale experiments from the literature by measuring the error for each model. They concluded that both models: Genova et al. and Casal give good results with less error. For this reason and because it is the most recent, the model of Genova et al. will be used in this article.

Indeed, Genova et al. proposed a new model to estimate the expansion energy, assuming that the explosion energy is mainly due to the flashing of liquid that can be observed as a thermal process driven by the excess heat stored in the liquid itself. This expansion energy can be expressed as:

$$E = m_l C_p (T_{rupt} - T_b) = m_l C_p \Delta T \tag{12}$$

where m_l is the liquid mass, C_p is the specific heat of the liquid at constant pressure (C_p variation as a function of temperature may be neglected), T_{rupt} is the temperature of the overheated liquid and T_b is the boiling temperature of the liquid at the atmospheric pressure.

The resulting expansion energy is then multiplied by a factor two for ground effect and used in Sedov-Taylor model to calculate the blast wave overpressure.

In this paper, the published works of Reid et al. (1987), Younglave and Ely (1987), Bücker and Wagner (2006) and Lemmon et al. (2009) are used as databases for the physico-chemical properties of LPG pure substances (such as boiling temperature, specific heat, etc.). Thus, the temperature of overheated liquid is extracted from the liquid-vapor equilibrium curves, knowing the burst pressure.

2.4. TNT equivalency model

The TNT equivalency model is one of the easiest to use models and has been widely applied for QRA studies. It is based on the assumption of an equivalence between the flammable substance and TNT, factored by an explosion efficiency term:

$$W_{TNT} = \frac{\eta M \Delta H_C}{E_{TNT}} \tag{13}$$

where W_{TNT} is the equivalent mass of TNT (kg), η is an empirical explosion efficiency (a value of 3% is recommended), M is the mass of fuel (kg), ΔH_C is the heat of combustion of fuel (kJ/kg) and E_{TNT} is the heat of combustion of TNT (4700 kJ/kg).

The equivalent mass of TNT is then used in the following equation to determine the scaling factor Z which is used to determine the scaled overpressure P_s/P_0:

$$Z = \frac{R}{W_{TNT}^{1/3}} \tag{14}$$

where R is the explosion distance (m).

Table 2
Summary of British Gas tests data.

Test No	Fluid	m_l(kg)	Burst pressure (bar)	T_{rupt} (K)	T_b (K)	C_p (J/(kg K))
1	Butane	2000	14.6	370	272	3055
2	Butane	2000	15.1	372	272	3088
3	Butane	1000	15.2	372	272	3088
4	Butane	2000	7.7	342	272	2746
5	Butane	2000	15.1	372	272	3088
6	Propane	2000	15.2	318	231	2995
7	Butane	2000	15.2	372	272	3088

The overpressure vs. distance can by calculated using the following empirical relation from Kinney and Graham (Mannan, 2012) for high explosives:

$$\frac{P_s}{P_0} = \frac{808\left[1+\left(\frac{Z}{4.5}\right)^2\right]}{\sqrt{\left[1+\left(\frac{Z}{0.048}\right)^2\right]}\sqrt{\left[1+\left(\frac{Z}{0.32}\right)^2\right]}\sqrt{\left[1+\left(\frac{Z}{1.35}\right)^2\right]}} \tag{15}$$

In the next Section, the TNT equivalent model will be used for comparison with the Sedov-Taylor model predictions.

3. Validation and discussion

To validate the proposed model, experimental data of BLEVE overpressure measurements are needed. Nevertheless, few experimental results are available in the literature and their reports are not always accessible. Thus, our validation will be limited to important large and medium-scale series of BLEVE tests, namely: British Gas tests series (Johnson et al., 1990) and Birk et al. (2007) experiments.

3.1. Large-scale experiments

In 1990, Johnson et al. conducted a series of seven BLEVE tests with horizontal cylindrical tanks of 5.6 m^3 of capacity, containing 2 tons of commercial butane that was heated until a pressure of 15.2 bar. The tanks were thermally insulated and their contents were heated by internal electrical resistances. The rupture of the reservoirs was triggered by the detonation of an explosive charge. A parametric analysis of the fluid mass (changed to 1 ton), tank volume (doubled), fuel type (changed to propane) and rupture pressure (halved) was performed. A brief summary of the data considered in these tests are reported in Table 2.

Fig. 2 presents comparisons of the overpressure evolution between the Sedov-Taylor model and the TNT equivalent model, based on the measurements of the seven British Gas tests. As shown in these figures, the prediction of Sedov-Taylor model compared to these tests measurements from the side of the reservoir gives a good agreement with slight error that may be considered as acceptable at the industrial scale. Indeed, the proposed model gives values close to the experimental measurements while the TNT equivalent model gives overestimated values (despite using a low explosion efficiency in the order of 3%).

Table 3 gives an estimate of the relative error related to the overpressure prediction by comparing Sedov-Taylor model with TNT equivalent model and the other models studied by Laboureur et al. (2014), based on the experimental measurements. This relative error is estimated for the different overpressures measured using Eq. (16). The average relative error of the four measurement points is calculated for each test as well as for all the tests. The

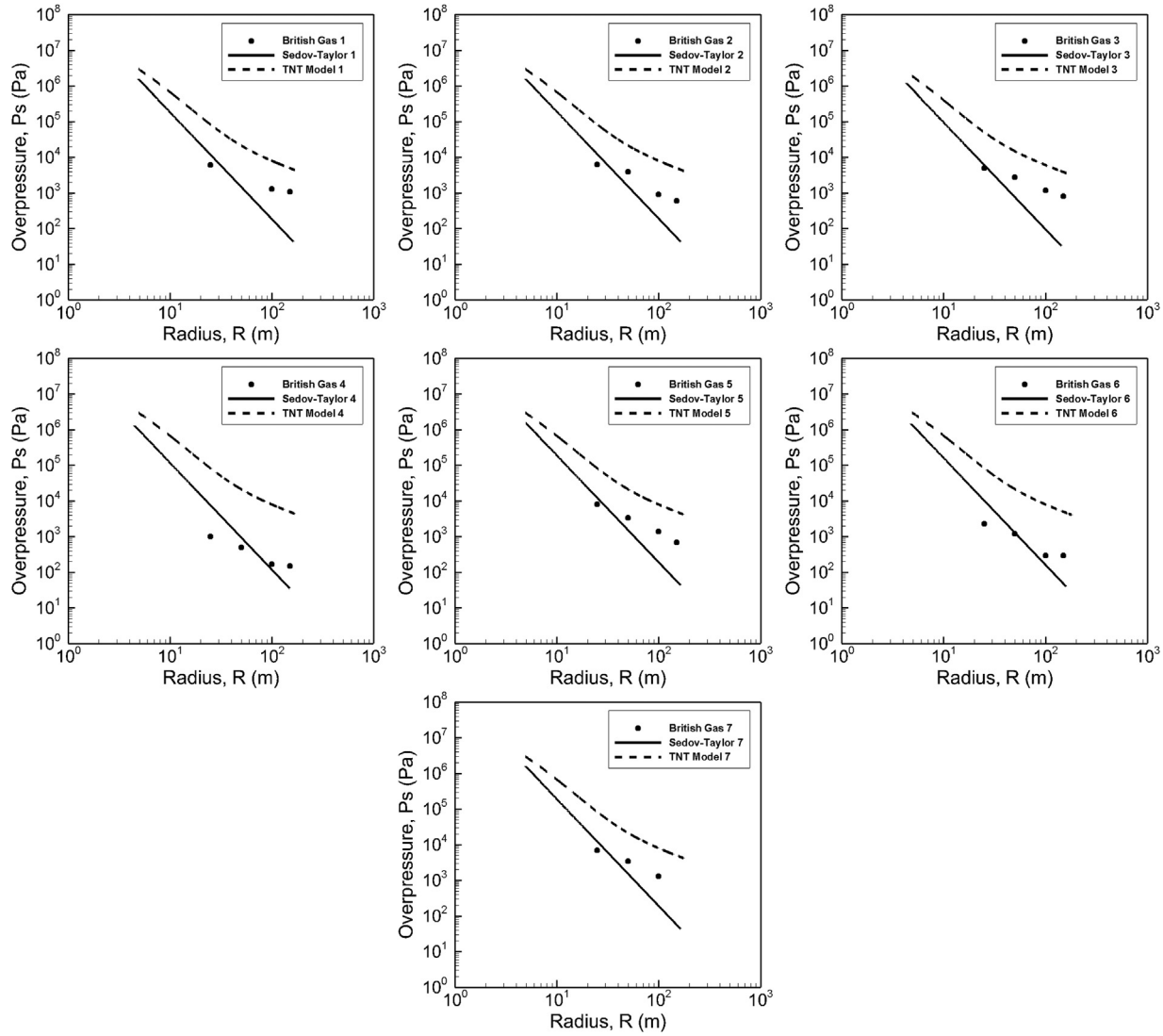

Fig. 2. Comparison of Sedov-Taylor and TNT equivalent predictions with British Gas experiments.

Table 3
Average relative error estimation for different models with large-scale tests.

Experiment	Average relative error of Sedov-Taylor prediction	Average relative error of TNT Equivalency model	Relative error calculated by Laboureur et al.							
			Prugh	Casal IS (*)	Casal IR (*)	Planas	Roberts	Genova	TNO	Birk
British Gas 1	−0.30	6.99	0.60	0.91	0.23	0.17	0.71	2.64	0.71	−0.26
British Gas 2	−0.34	7.93	1.01	1.33	0.50	0.47	1.16	0.30	1.16	−0.06
British Gas 3	−0.60	5.40	0.88	1.06	0.35	0.37	0.99	0.14	0.99	−0.34
British Gas 4	1.60	50.46	8.64	10.57	6.56	6.50	9.16	5.58	9.15	3.27
British Gas 5	−0.46	6.28	0.92	1.01	0.31	0.41	1.02	0.15	1.02	0.45
British Gas 6	0.57	23.38	3.73	2.72	1.40	2.38	3.97	1.99	3.81	1.13
British Gas 7	−0.22	7.08	0.65	0.95	0.26	0.19	0.71	0.02	0.70	−0.27
All	0.04	15.36	2.45	2.75	1.44	1.58	2.64	1.24	2.62	0.61

(*) IS and IR stand for Casal et al. models with isentropic and irreversible expansions, respectively.

results in Table 3 show that the proposed model gives the best prediction with the least error.

$$Relative\ error = \frac{Model\ prediction - experimental\ measurement}{experimental\ measurement} \quad (16)$$

Often, it is also useful to know the relationship between the results issued from an experiment and those given by a prediction model. This is easily carried out by a quantitative evaluation of the correlation using Pearson's correlation coefficient, which represents a statistical measure of the strength of a linear relationship between paired data. This coefficient, denoted r_{co}, writes as:

$$r_{co} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\left(\sum (x_i - \bar{x})^2\right)\left(\sum (y_i - \bar{y})^2\right)}} \quad (17)$$

where x_i is a measured value, $\bar{x}$ is the mean of x_i values, y_i is a predicted value and $\bar{y}$is the mean of y_i values.

Table 4
Estimation of correlation coefficient based on large-scale tests.

x_i values source	y_i values source	Coefficient r_{co}	Strength
British Gas test 1	Sedov-Taylor model 1	0.99	Very strong
British Gas test 2	Sedov-Taylor model 2	0.89	Very strong
British Gas test 3	Sedov-Taylor model 3	0.94	Very strong
British Gas test 4	Sedov-Taylor model 4	0.95	Very strong
British Gas test 5	Sedov-Taylor model 5	0.97	Very strong
British Gas test 6	Sedov-Taylor model 6	0.94	Very strong
British Gas test 7	Sedov-Taylor model 7	0.96	Very strong
All		0.95	Very strong

Note that r_{co} values belong to [−1, +1] and can be interpreted as:

- Positive values indicate proportional correlation.
- Negative values indicate an inverse correlation.
- A value of 1 or −1, corresponds to a perfect correlation.
- A value of 0 indicates no relationship between the variables.

Correlation strength can be interpreted verbally using the famous classification of Evans for the absolute value of r_{co} (Evans, 1996):

- 0.00–0.19: very weak
- 0.20–0.39: weak
- 0.40–0.59: moderate
- 0.60–0.79: strong
- 0.80–1.00: very strong

In this case, to better analyze our results, we have calculated for each test an estimate of the correlation coefficient between our predicted overpressure values and the measured overpressure values. The average correlation coefficient is presented for all the tests, as shown in Table 4. The average correlation coefficient value of 0.95 shows that there is a very strong positive correlation between the Sedov-Taylor model prediction and the experimental measurements. This result is in concordance with comparison results of Fig. 2.

From the previous results, it is clear that Sedov-Taylor model gives good predictions in the near field where overpressure is important. Otherwise, in 17 of the 26 experimental presented results (65% of the events), it underestimates the overpressure obtained as a function of the radius in the far field when the overpressure becomes low (with a gap which does not exceed almost 20 mbar).

Table 5
Summary of Birk et al. experiments data.

Test No	Fluid	m_l(kg)	Burst pressure (bar)	T_{rupt} (K)	T_b (K)	C_p (J/(kg K))
1	Propane	150	18.63	327	231	3201
2	Propane	309.4	18.46	328	231	3201
3	Propane	116.9	16.99	328	231	3201
4	Propane	184.6	18.94	330	231	3337
5	Propane	109.5	15.73	322	231	3089
6	Propane	453.1	18.03	330	231	3337
7	Propane	475.2	15.63	320	231	3089
8	Propane	470.3	18.13	325	231	3089
9	Propane	538.5	18.58	327	231	3201

3.2. Medium-scale experiments

Between 2000 and 2004, Birk et al. conducted a series of BLEVE tests with ASME code propane tanks of 2000 liters of design capacity were engulfed by fire. Among the series of tests conducted, nine of them resulted in a BLEVE, with a failure pressure between 15 bar and 19 bar (Birk et al., 2007). The conditions and results of these tests carried out are reported in Table 5.

The comparison of the overpressure evolution between Sedov-Taylor model and TNT equivalent model, based on the nine Birk et al. experiments, is presented in Fig. 3. These figures show that the predictions of Sedov-Taylor model are in good agreement with the experimental data (obtained from the axial direction of tanks). Indeed, this model gives predictions close to the measurements while the TNT equivalent model overestimes the overpressure values.

Table 6 gives an estimate of the relative error related to overpressure prediction by comparing Sedov-Taylor model to TNT equivalent model and other models studied by Laboureur et al. (2014), based on the previously presented measurements. From Table 6, the results show that Sedov-Taylor approach gives predictions similar to those obtained by the models of Casal IS, Planas and Birk. It also gives better predictions by comparing to the results issued from Prugh, Roberts and TNO models.

Through the calculation of the mean correlation coefficient value, it is shown that there is also a very strong positive correlation between the Sedov-Taylor prediction and the measurements obtained from the medium-scale tests, as shown in Table 7.

In 19 of the 33 medium-scale measurements (58% of the events), Sedov-Taylor model underestimates the overpressure in the far field with a gap that does not generally exceed 20 mbar, in most observed cases. On the other hand, it gives good predictions in the near field.

In addition, the results issued from the various studied experiments show that the Sedov-Taylor approcah is more suitable for large-scale explosions than medium-scale explosions. Indeed, the

Table 6
Average relative error estimation for different models with medium-scale tests.

Experiment	Average relative error of Sedov-Taylor prediction	Average relative error of TNT Equivalency model	Relative error calculated by Laboureur et al.							
			Prugh	Casal IS (*)	Casal IR (*)	Planas	Roberts	Genova	TNO	Birk
Birk et al. test 1	−0.27	6.24	0.77	0.14	−0.26	0.22	0.87	−0.19	0.84	0.76
Birk et al. test 2	1.32	15.64	1.97	1.20	0.40	1.03	2.17	0.54	2.12	1.23
Birk et al. test 3	−0.09	7.91	1.41	0.44	−0.07	0.66	1.53	0.05	1.46	1.49
Birk et al. test 4	0.97	12.74	1.97	0.95	0.22	0.98	2.19	0.33	2.13	1.77
Birk et al. test 5	−0.22	8.58	1.75	0.60	0.03	0.91	1.80	0.21	1.71	1.86
Birk et al. test 6	0.17	6.46	0.23	−0.03	−0.39	−0.16	0.28	−0.32	0.24	−0.27
Birk et al. test 7	1.88	20.27	2.22	1.56	0.62	1.28	2.39	0.91	2.31	0.94
Birk et al. test 8	2.42	24.70	2.09	1.57	0.61	1.16	2.32	0.80	2.22	0.93
Birk et al. test 9	1.88	16.27	1.45	1.10	0.31	0.73	1.70	0.44	1.64	0.41
All	0.89	13.20	1.52	0.84	0.17	0.74	1.67	0.31	1.60	0.95

(*) IS and IR stand for Casal et al. models with isentropic and irreversible expansions, respectively.

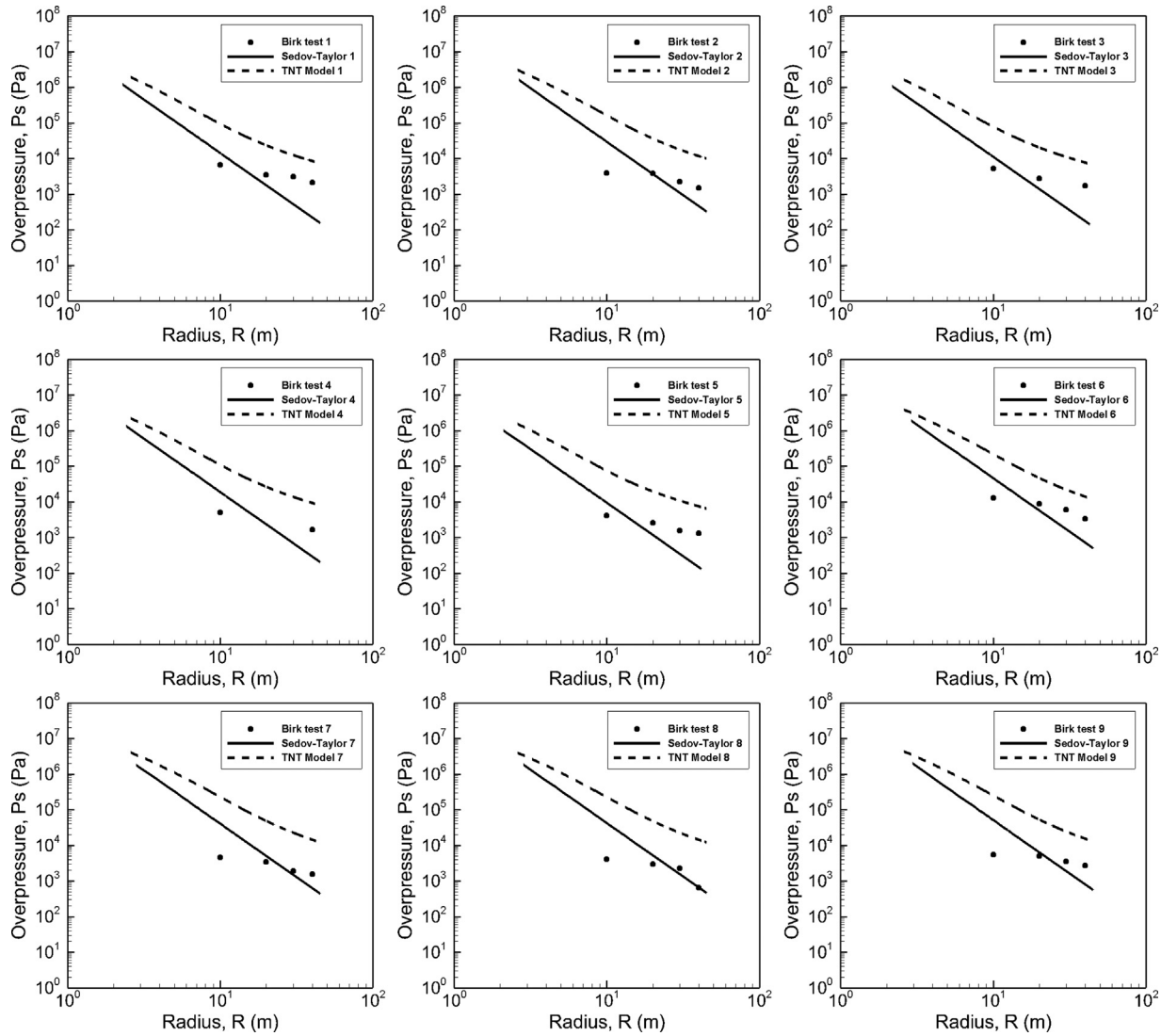

Fig. 3. Comparison of Sedov-Taylor and TNT equivalent predictions with Birk et al. experiments.

Table 7
Estimation of correlation coefficient based on medium-scale tests.

x_i values source	y_i values source	Coefficient r_{co}	Strength
Birk et al. test 1	Sedov-Taylor model 1	0.97	Very strong
Birk et al. test 2	Sedov-Taylor model 2	0.68	Strong
Birk et al. test 3	Sedov-Taylor model 3	0.98	Very strong
Birk et al. test 4	Sedov-Taylor model 4	1.00	Very strong
Birk et al. test 5	Sedov-Taylor model 5	0.94	Very strong
Birk et al. test 6	Sedov-Taylor model 6	0.89	Very strong
Birk et al. test 7	Sedov-Taylor model 7	0.87	Very strong
Birk et al. test 8	Sedov-Taylor model 8	0.79	Strong
Birk et al. test 9	Sedov-Taylor model 9	0.72	Strong
All		0.87	Very strong

greater the explosion magnitude, the more accurate the Sedov-Taylor prediction (Sedov, 1959; Taylor, 1950a,b).

The health and safety community always opts for a conservative approach (taking into account the worst-case scenario), because it is better to make an error by overprotecting than by underprotecting the equipment and people from the design stage. For these reasons, it is better to take into account an additional safety margin in the order of 20 mbar when the overpressure predicted by Sedov-Taylor model becomes weak in the far field.

4. Conclusion and future work

In this article (Part I), we proposed an analytical model based on the self-similar theory, which allows the estimate of the overpressure effect generated by BLEVE phenomenon. Known as a powerful approach, dimensional analysis based on the Pi theorem has been used to determine the different physical quantities characterizing the blast wave evolution (radius, velocity and overpressure).

To demonstrate the great interest of the proposed model, a validation with large and medium-scale BLEVE experiments was carried out. Moreover, a comparison with the TNT equivalent model and other models (empirical and simplified physical) was performed. In terms of prediction precision, the results of this validation show a good agreement with slight error that can be considered acceptable at the industrial scale. Thus, a very strong positive correlation was found between Sedov-Taylor prediction and experimental data. The comparison showed that the proposed model gives good predictions in the near field where the overpressure is sufficiently large and it gives underestimates in the far field when the overpressure becomes weak with a small gap. It has also been shown that Sedov-Taylor model is more suitable for large-scale explosions than medium-scale explosions.

The results obtained in this Part I can be considered as a first step. We believe that this analytical model could be successfully applied in Quantitative Risk Analysis (QRA) as a tool for estimating the explosion blast effects. Indeed, the integration of the model into the QRA would be an opportunity in the risk management, allowing a good characterization of potential hazardous scenarios, and consequently a better risk decision-making.

Application of Sedov-Taylor model to an Algerian gas processing unit is discussed in a companion article (Part II), where QRA is chosen as a methodological framework.

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